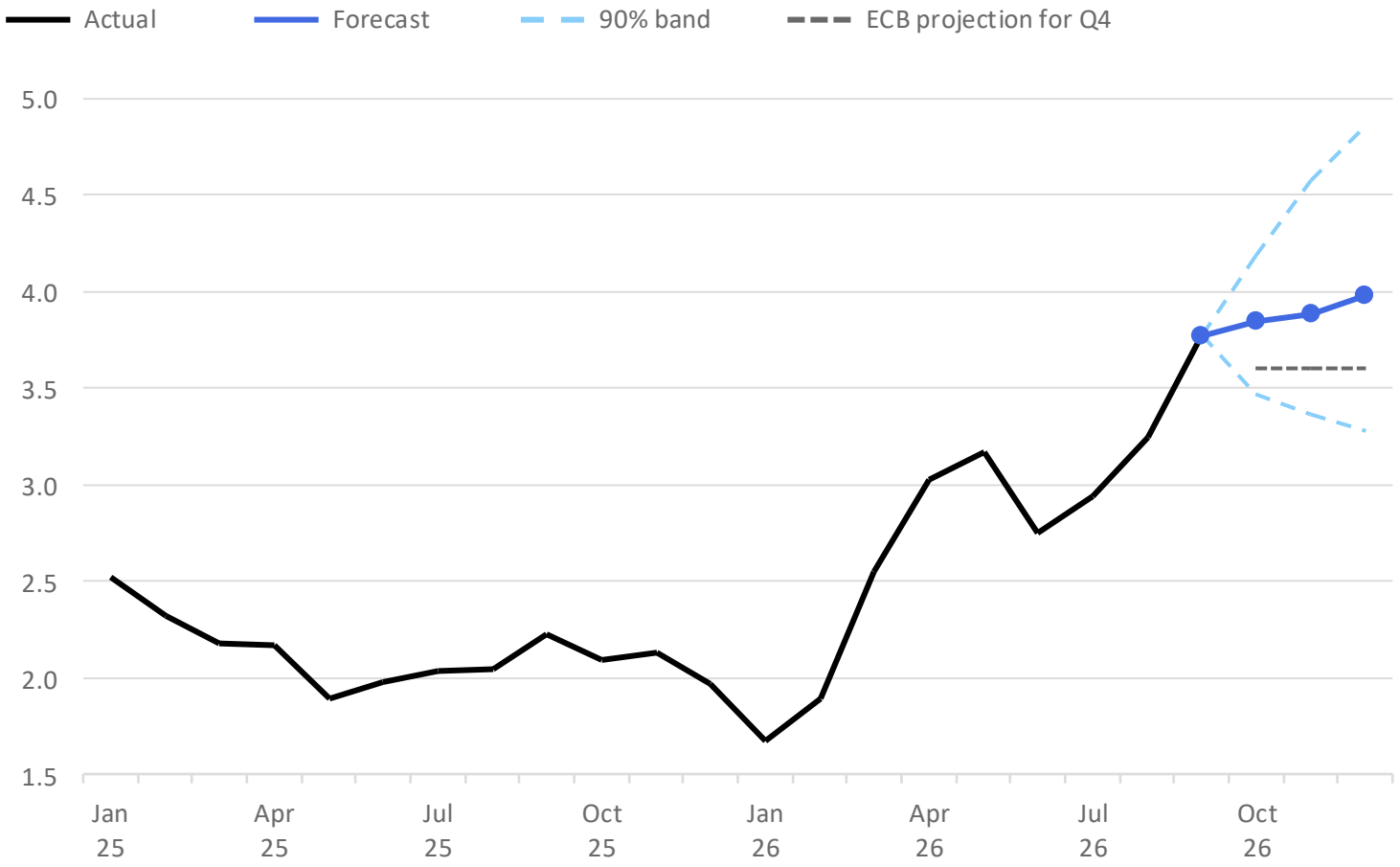
Inflation should stay close to 4% until the end of the year

Euro area HICP, annual rate, %. Forecast made on 20 October 2026 with data to September.



3.85% October
90% band 3.5 to 4.2

3.88% November
90% band 3.4 to 4.6

3.98% December
90% band 3.3 to 4.9

From September's 3.77%, the year-ago months dropping out take 0.11 pp off December and the new months add 0.32.

The ECB's September projection has 3.6% for Q4, with oil at about $88. Brent averaged $127 in the first half of October.

To land on the ECB's 3.6%, the same model needs Brent near **$81**. At $90 it gives 3.67%, so most of the gap is the oil price.

Sources: Eurostat, ECB staff projections September 2026, FRED. The band comes from the model's own past errors.

Rebuilt from its parts, the headline matches Eurostat to within 0.04 pp

A pipeline from the Eurostat API that checks itself before anything is modelled.

From the API to the dashboard

Pull the 5 HICP tables and 3 business surveys from the Eurostat API, plus Brent, EUR/USD, gas and weekly pump prices.
Keep every raw response as it came back, with a new copy only when the source changes.
Check the data, 17 checks in all. One of them holds each year's country weights against who was in the euro, from 11 countries in 1999 to 21 in 2026. Then rebuild the headline from its parts; any failure stops the run.
Use the same tables for the forecast, the shock engine and the dashboard. One command runs it all: python -m src all

The headline rebuilt from its parts

Euro area rebuilt from	months	largest gap, pp
the 6 main groups	298	0.030
the 13 COICOP divisions	346	0.019
the 21 country indices	346	0.038
groups within each country	298	0.032
divisions within each country	346	0.026
group contributions, vs Eurostat's	297	0.009

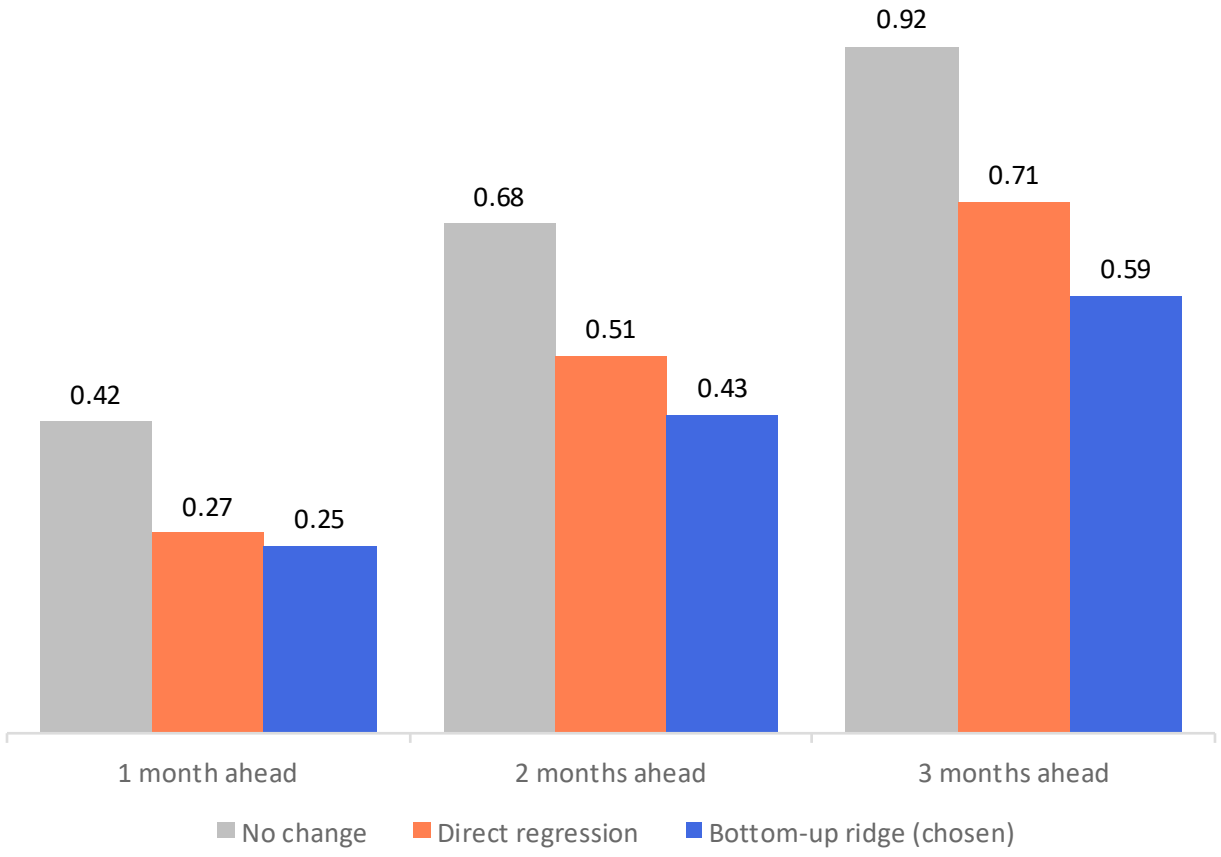
Limit 0.05 pp, 0.02 for contributions. What's left is rounding: official indices have two decimals.

Worth knowing about the data

- Eurostat's old HICP tables stop at December 2025. Everything here uses the new ECOICOP ver.2 ones.
- Headline rates are not revised after the final release: all 321 months since 2000 match today's figures.
- Weekly pump prices from the Commission's Oil Bulletin explain 97% of monthly moves in HICP fuels.

In a real-time backtest the forecast beats 'no change' by about 40%

Forecast error (RMSE), pp of the annual rate. 138 forecasts made between 2015 and 2026.



Same months for every model, scored against today's data; against first releases the ranking is the same. The plain bottom-up model, not shown, is close behind (0.25, 0.47, 0.64) and the gap is not significant.

How it works

I forecast next month's price change for each of the 6 main groups and chain them into the headline. The year-ago months dropping out of the annual rate are already known, so only the new months are forecast.

How it was tested

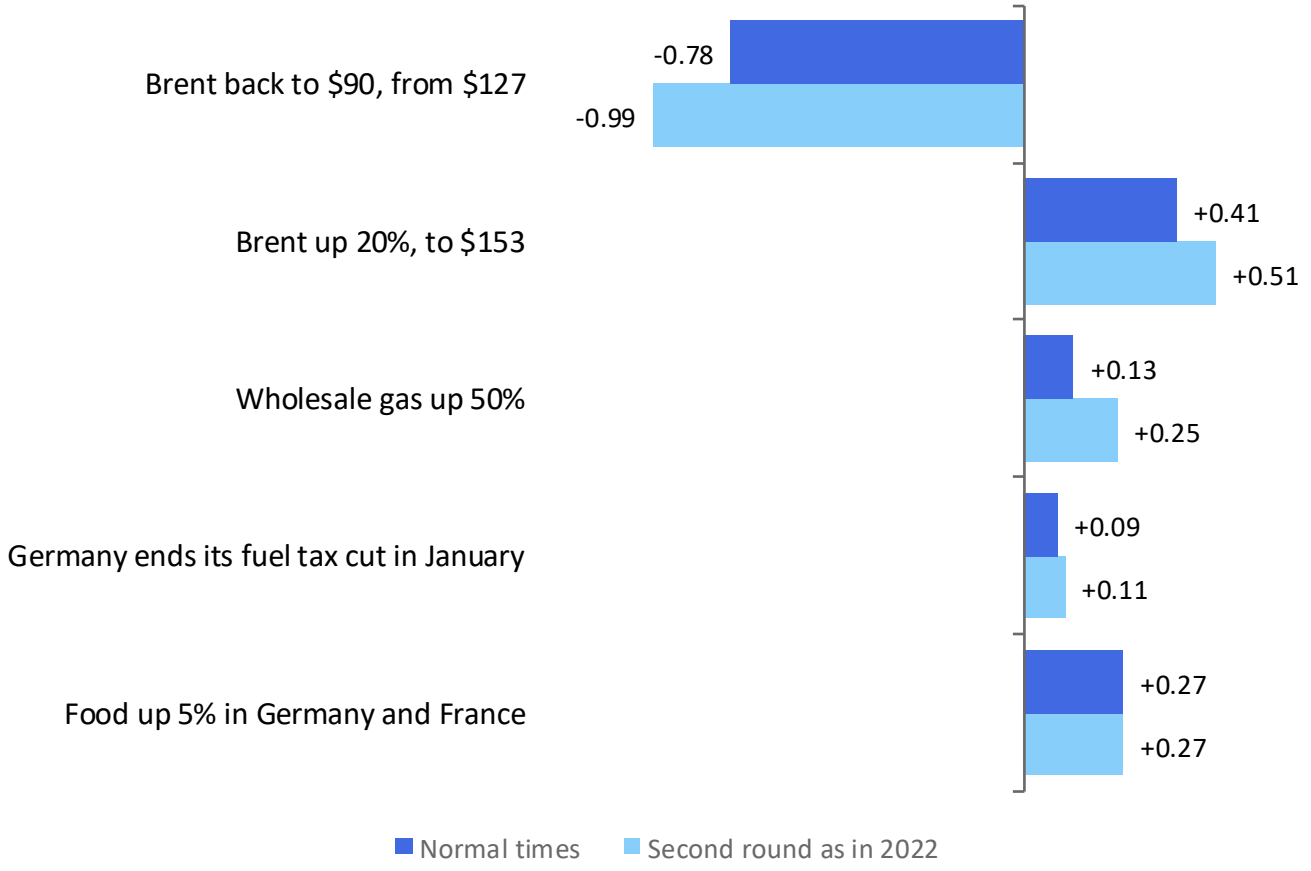
- Each forecast uses only what was published that day, and a test with poisoned future data makes sure of it.
- Six models, simplest first, and a rule set before the backtest: lowest average error, ties within 2% go to the simpler one.
- Picked with 2015-19 data only, the same model still came first on 2020-26.
- Taking out the business surveys raises the error by 11%, taking out weekly pump prices by 8%. Gas doesn't help.

Where it's weak

The 90% band caught 88 to 94% of outcomes outside 2021-23, but only 56 to 67% during the surge.

Oil back at $90 would take about 0.8 pp off inflation in 2027

Change in the euro area annual rate, 2027 average, pp. Shocks start in November 2026 unless stated.



Germany's cut: 16.7 cents a litre on 5 October, read off the Weekly Oil Bulletin. Germany is 27.7% of the euro area basket.

How the simulator works

- Shock any of the 6 groups or 13 divisions, in any of the 21 countries, and choose how it moves over time.
- The direct effect is exact. It runs through each country's and group's weight and Eurostat's December chain-linking, and matches a full rebuild of the index.
- Oil and gas reach energy prices, then energy feeds food, goods and services over the next year.

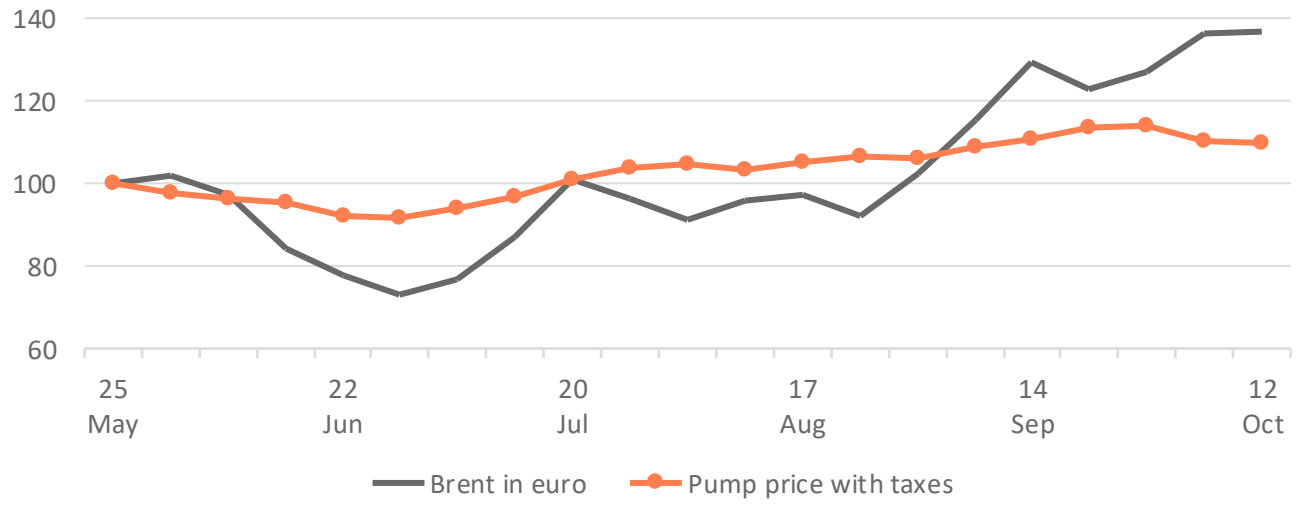
What I found

- One group in one country works too: German fuel prices up 10% add 0.13 pp to the euro area rate in 2027, 0.02 of it through the second round.
- Brent up 10% for good lifts the price level by about 0.3% after a year in normal times, 0.4% with 2020-23 in the sample.
- The second round is three times stronger with 2020-23 in the sample: processed food moves 0.17% per 1% on energy, against 0.06% without.

Limits, and what I'd do next

Code: https://github.com/hoangdatkt1107/euro_area_hicp. Dashboard: https://euro-area-inflation.streamlit.app/ (if it's asleep, it takes a minute to wake up)

October: pump prices fell while Brent rose



Petrol and diesel with taxes, euro area average (Weekly Oil Bulletin). Brent in euro (FRED, ECB). 25 May = 100.

Check it on 4 November

My October forecast, saved on 20 October: 3.85%, 90% band 3.5 to 4.2. Eurostat's flash estimate comes out on 4 November.

Limits

- The bands come from past errors, so they were far too narrow in 2021-23.
- October fuels: pump prices alone say -1.7%, the model +0.1% as it also weighs Brent and the usual October pattern. About 0.1 pp on the headline.
- No spillover between countries, no wage or wider exchange-rate channel, and 2027 weights are taken equal to 2026.

Next

- Nowcast with the German, French, Italian and Spanish flash CPIs, which come out just before Eurostat's. Without them the model probably trails the consensus one month ahead.
- Oil futures and inflation swaps for the energy path.
- Quantile regression forests through the same backtest, so the bands can widen when volatility jumps.